

```
def karatsuba(x, y):
    if x < 10 or y < 10:
        return x * y

    n = max(len(str(x)), len(str(y)))
    m = n // 2

    high1 = x // (10 ** m)
    low1 = x % (10 ** m)

    high2 = y // (10 ** m)
    low2 = y % (10 ** m)

    z0 = karatsuba(low1, low2)
    z1 = karatsuba(low1 + high1, low2 + high2)
    z2 = karatsuba(high1, high2)

    return z2 * (10 ** (2 * m)) + \
        (z1 - z2 - z0) * (10 ** m) + z0

x = int(input("Enter first number: "))
y = int(input("Enter second number: "))

print("Product:", karatsuba(x, y))
```

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:40)
on win32
Enter "help" below or click "Help" above for more info

>>>
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Docu
Enter first number: 1234
Enter second number: 5678
Product: 7006652
```

```
>>>
```